



ELMALG: Rational Expressions and Quadratics

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Learning Objectives

- Find the domain of a rational expression (excluded values)
- Solve quadratic equations by factoring
- Apply the quadratic formula to find roots
- Interpret the discriminant to describe the nature of roots

Simplify each expression completely. Show all steps and circle your final answer.

Factoring trinomials

1. Factor completely: $x^2 + 4x + 4$

$$x^2 + 4x + 4$$

Answer: _____

2. The area of a rectangular garden is $x^2 + 11x + 24$ sq ft. Factor to find expressions for the length and width.

$$x^2 + 11x + 24$$

Answer: _____

3. Factor completely: $x^2 + 9x + 14$

$$x^2 + 9x + 14$$

Answer: _____

4. The area of a rectangular garden is $x^2 + 18x + 80$ sq ft. Factor to find expressions for the length and width.

$$x^2 + 18x + 80$$

Answer: _____

5. Factor completely: $x^2 + 11x + 18$

$$x^2 + 11x + 18$$

Answer: _____



6. The area of a rectangular garden is $x^2 + 12x + 27$ sq ft. Factor to find expressions for the length and width.

$$x^2 + 12x + 27$$

Answer: _____

7. Factor completely: $x^2 + 17x + 72$

$$x^2 + 17x + 72$$

Answer: _____

8. The area of a rectangular garden is $x^2 + 15x + 50$ sq ft. Factor to find expressions for the length and width.

$$x^2 + 15x + 50$$

Answer: _____

9. Factor completely: $x^2 + 8x + 12$

$$x^2 + 8x + 12$$

Answer: _____

10. The area of a rectangular garden is $x^2 + 20x + 96$ sq ft. Factor to find expressions for the length and width.

$$x^2 + 20x + 96$$

Answer: _____

Solving quadratic equations

11. Solve by factoring: $1x^2 + -6x + 8 = 0$.

$$x^2 - 6x + 8 = 0$$

Answer: _____

12. Find the x-intercepts of $f(x) = 1x^2 + -1x + -12$ by solving the equation $= 0$.

$$x^2 - 1x - 12 = 0$$

Answer: _____

13. Solve by factoring: $x^2 + 2x + -24 = 0$

$$x^2 + 2x - 24 = 0$$

Answer: _____



14. Solve by factoring: $1x^2 + -3x + -4 = 0$.

$$x^2 - 3x - 4 = 0$$

Answer: _____

15. Find the x-intercepts of $f(x) = 1x^2 + 2x + -3$ by solving the equation $= 0$.

$$x^2 + 2x - 3 = 0$$

Answer: _____

16. Solve by factoring: $x^2 + 0x + -4 = 0$

$$x^2 - 4 = 0$$

Answer: _____

17. Solve by factoring: $1x^2 + 2x + 1 = 0$.

$$x^2 + 2x + 1 = 0$$

Answer: _____

18. Find the x-intercepts of $f(x) = 1x^2 + 3x + -4$ by solving the equation $= 0$.

$$x^2 + 3x - 4 = 0$$

Answer: _____

19. Solve by factoring: $x^2 + 1x + -12 = 0$

$$x^2 + 1x - 12 = 0$$

Answer: _____

20. Solve by factoring: $2x^2 + -4x + 0 = 0$.

$$2x^2 - 4x = 0$$

Answer: _____

21. Find the x-intercepts of $f(x) = 1x^2 + 2x + -15$ by solving the equation $= 0$.

$$x^2 + 2x - 15 = 0$$

Answer: _____

22. Solve by factoring: $x^2 + 5x + -6 = 0$

$$x^2 + 5x - 6 = 0$$

Answer: _____

Domain of rational expressions



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23. Find the value of x that makes the expression undefined: $f(x) = (3x + -4) / (4x + 6)$.

$$f(x) = \frac{3x + -4}{4x + 6}$$

Answer: _____

24. Find the value of x that makes the expression undefined: $f(x) = (2x + 2) / (3x + 7)$.

$$f(x) = \frac{2x + 2}{3x + 7}$$

Answer: _____

25. Find the value of x that makes the expression undefined: $f(x) = (4x + -3) / (2x + 2)$.

$$f(x) = \frac{4x + -3}{2x + 2}$$

Answer: _____

26. Find the value of x that makes the expression undefined: $f(x) = (4x + 3) / (1x + -1)$.

$$f(x) = \frac{4x + 3}{1x + -1}$$

Answer: _____

Simplifying rational expressions

27. Simplify the rational expression: $(4x + 8) / (4x + 24)$. State any restrictions on x .

$$\frac{4x + 8}{4x + 24}$$

Answer: _____

28. Simplify the rational expression: $(2x + 4) / (2x + 6)$. State any restrictions on x .

$$\frac{2x + 4}{2x + 6}$$

Answer: _____

29. Simplify the rational expression: $(3x + 3) / (3x + 24)$. State any restrictions on x .

$$\frac{3x + 3}{3x + 24}$$

Answer: _____



30. Simplify the rational expression: $(2x + 2) / (2x + 16)$. State any restrictions on x .

$$\frac{2x + 2}{2x + 16}$$

Answer: _____





Topics: Domain of rational expressions, Simplifying rational expressions, Factoring trinomials, Solving quadratic equations. All answers verified by independent computation.

Solutions



Factoring trinomials

1. Factor completely: $x^2 + 4x + 4$

$$x^2 + 4x + 4$$

→ Find two integers that multiply to 4 and add to 4: they are 2 and 2.

→ Factored form: $(x + 2)(x + 2)$.

Answer: $(x + 2)(x + 2)$

2. The area of a rectangular garden is $x^2 + 11x + 24$ sq ft. Factor to find expressions for the length and width.

$$x^2 + 11x + 24$$

→ Find two integers that multiply to 24 and add to 11: they are 3 and 8.

→ Length and width: $(x + 3)(x + 8)$.

Answer: $(x + 3)(x + 8)$

3. Factor completely: $x^2 + 9x + 14$

$$x^2 + 9x + 14$$

→ Find two integers that multiply to 14 and add to 9: they are 2 and 7.

→ Factored form: $(x + 2)(x + 7)$.

Answer: $(x + 2)(x + 7)$

4. The area of a rectangular garden is $x^2 + 18x + 80$ sq ft. Factor to find expressions for the length and width.

$$x^2 + 18x + 80$$

→ Find two integers that multiply to 80 and add to 18: they are 10 and 8.

→ Length and width: $(x + 10)(x + 8)$.

Answer: $(x + 10)(x + 8)$

5. Factor completely: $x^2 + 11x + 18$

$$x^2 + 11x + 18$$

→ Find two integers that multiply to 18 and add to 11: they are 9 and 2.

→ Factored form: $(x + 9)(x + 2)$.

Answer: $(x + 9)(x + 2)$

6. The area of a rectangular garden is $x^2 + 12x + 27$ sq ft. Factor to find expressions for the length and width.

$$x^2 + 12x + 27$$

→ Find two integers that multiply to 27 and add to 12: they are 3 and 9.

→ Length and width: $(x + 3)(x + 9)$.

Answer: $(x + 3)(x + 9)$



7. Factor completely: $x^2 + 17x + 72$

$$x^2 + 17x + 72$$

→ Find two integers that multiply to 72 and add to 17: they are 8 and 9.

→ Factored form: $(x + 8)(x + 9)$.

Answer: $(x + 8)(x + 9)$

8. The area of a rectangular garden is $x^2 + 15x + 50$ sq ft. Factor to find expressions for the length and width.

$$x^2 + 15x + 50$$

→ Find two integers that multiply to 50 and add to 15: they are 10 and 5.

→ Length and width: $(x + 10)(x + 5)$.

Answer: $(x + 10)(x + 5)$

9. Factor completely: $x^2 + 8x + 12$

$$x^2 + 8x + 12$$

→ Find two integers that multiply to 12 and add to 8: they are 2 and 6.

→ Factored form: $(x + 2)(x + 6)$.

Answer: $(x + 2)(x + 6)$

10. The area of a rectangular garden is $x^2 + 20x + 96$ sq ft. Factor to find expressions for the length and width.

$$x^2 + 20x + 96$$

→ Find two integers that multiply to 96 and add to 20: they are 8 and 12.

→ Length and width: $(x + 8)(x + 12)$.

Answer: $(x + 8)(x + 12)$



Solving quadratic equations

11. Solve by factoring: $1x^2 + -6x + 8 = 0$.

$$x^2 - 6x + 8 = 0$$

→ Factor the quadratic. Set each factor = 0.

→ Solutions: $x = 2$ and $x = 4$.

Answer: $x = 2$ or $x = 4$

12. Find the x-intercepts of $f(x) = 1x^2 + -1x + -12$ by solving the equation = 0.

$$x^2 - 1x - 12 = 0$$

→ Set $1x^2 + -1x + -12 = 0$. Factor or use the quadratic formula.

→ $x = -3$ or $x = 4$.

Answer: $x = -3$ or $x = 4$

13. Solve by factoring: $x^2 + 2x + -24 = 0$

$$x^2 + 2x - 24 = 0$$

→ Factor: find two numbers that multiply to -24 and add to 2.

→ Set each factor equal to zero.

→ $x = -6$ or $x = 4$.

Answer: $x = -6$ or $x = 4$

14. Solve by factoring: $1x^2 + -3x + -4 = 0$.

$$x^2 - 3x - 4 = 0$$

→ Factor the quadratic. Set each factor = 0.

→ Solutions: $x = -1$ and $x = 4$.

Answer: $x = -1$ or $x = 4$

15. Find the x-intercepts of $f(x) = 1x^2 + 2x + -3$ by solving the equation = 0.

$$x^2 + 2x - 3 = 0$$

→ Set $1x^2 + 2x + -3 = 0$. Factor or use the quadratic formula.

→ $x = -3$ or $x = 1$.

Answer: $x = -3$ or $x = 1$

16. Solve by factoring: $x^2 + 0x + -4 = 0$

$$x^2 - 4 = 0$$

→ Factor: find two numbers that multiply to -4 and add to 0.

→ Set each factor equal to zero.

→ $x = -2$ or $x = 2$.

Answer: $x = -2$ or $x = 2$



17. Solve by factoring: $1x^2 + 2x + 1 = 0$.

$$x^2 + 2x + 1 = 0$$

→ Factor the quadratic. Set each factor = 0.

→ Solutions: $x = -1$ and $x = -1$.

Answer: $x = -1$ or $x = -1$

18. Find the x-intercepts of $f(x) = 1x^2 + 3x + -4$ by solving the equation = 0.

$$x^2 + 3x - 4 = 0$$

→ Set $1x^2 + 3x + -4 = 0$. Factor or use the quadratic formula.

→ $x = -4$ or $x = 1$.

Answer: $x = -4$ or $x = 1$

19. Solve by factoring: $x^2 + 1x + -12 = 0$

$$x^2 + 1x - 12 = 0$$

→ Factor: find two numbers that multiply to -12 and add to 1.

→ Set each factor equal to zero.

→ $x = -4$ or $x = 3$.

Answer: $x = -4$ or $x = 3$

20. Solve by factoring: $2x^2 + -4x + 0 = 0$.

$$2x^2 - 4x = 0$$

→ Factor the quadratic. Set each factor = 0.

→ Solutions: $x = 0$ and $x = 2$.

Answer: $x = 0$ or $x = 2$

21. Find the x-intercepts of $f(x) = 1x^2 + 2x + -15$ by solving the equation = 0.

$$x^2 + 2x - 15 = 0$$

→ Set $1x^2 + 2x + -15 = 0$. Factor or use the quadratic formula.

→ $x = -5$ or $x = 3$.

Answer: $x = -5$ or $x = 3$

22. Solve by factoring: $x^2 + 5x + -6 = 0$

$$x^2 + 5x - 6 = 0$$

→ Factor: find two numbers that multiply to -6 and add to 5.

→ Set each factor equal to zero.

→ $x = -6$ or $x = 1$.

Answer: $x = -6$ or $x = 1$



Domain of rational expressions

23. Find the value of x that makes the expression undefined: $f(x) = (3x + -4) / (4x + 6)$.

$$f(x) = \frac{3x + -4}{4x + 6}$$

→ Set denominator = 0: $4x + 6 = 0$.

→ $x = -3/2$ is excluded from the domain.

Answer: $x \neq -3/2$

24. Find the value of x that makes the expression undefined: $f(x) = (2x + 2) / (3x + 7)$.

$$f(x) = \frac{2x + 2}{3x + 7}$$

→ Set denominator = 0: $3x + 7 = 0$.

→ $x = -7/3$ is excluded from the domain.

Answer: $x \neq -7/3$

25. Find the value of x that makes the expression undefined: $f(x) = (4x + -3) / (2x + 2)$.

$$f(x) = \frac{4x + -3}{2x + 2}$$

→ Set denominator = 0: $2x + 2 = 0$.

→ $x = -1$ is excluded from the domain.

Answer: $x \neq -1$

26. Find the value of x that makes the expression undefined: $f(x) = (4x + 3) / (1x + -1)$.

$$f(x) = \frac{4x + 3}{1x + -1}$$

→ Set denominator = 0: $1x + -1 = 0$.

→ $x = 1$ is excluded from the domain.

Answer: $x \neq 1$



Simplifying rational expressions

27. Simplify the rational expression: $(4x + 8) / (4x + 24)$. State any restrictions on x .

$$\frac{4x + 8}{4x + 24}$$

→ Factor the numerator: $4(x + 2)$. Factor the denominator: $4(x + 6)$.

→ Cancel the common factor 4.

→ Simplified: $(x + 2)/(x + 6)$, where x is not equal to -6.

Answer: $\frac{x + 2}{x + 6}, x \neq -6$

28. Simplify the rational expression: $(2x + 4) / (2x + 6)$. State any restrictions on x .

$$\frac{2x + 4}{2x + 6}$$

→ Factor the numerator: $2(x + 2)$. Factor the denominator: $2(x + 3)$.

→ Cancel the common factor 2.

→ Simplified: $(x + 2)/(x + 3)$, where x is not equal to -3.

Answer: $\frac{x + 2}{x + 3}, x \neq -3$

29. Simplify the rational expression: $(3x + 3) / (3x + 24)$. State any restrictions on x .

$$\frac{3x + 3}{3x + 24}$$

→ Factor the numerator: $3(x + 1)$. Factor the denominator: $3(x + 8)$.

→ Cancel the common factor 3.

→ Simplified: $(x + 1)/(x + 8)$, where x is not equal to -8.

Answer: $\frac{x + 1}{x + 8}, x \neq -8$

30. Simplify the rational expression: $(2x + 2) / (2x + 16)$. State any restrictions on x .

$$\frac{2x + 2}{2x + 16}$$

→ Factor the numerator: $2(x + 1)$. Factor the denominator: $2(x + 8)$.

→ Cancel the common factor 2.

→ Simplified: $(x + 1)/(x + 8)$, where x is not equal to -8.

Answer: $\frac{x + 1}{x + 8}, x \neq -8$

